

Name _____

Section 2-2 – Inductive and Deductive Reasoning

VOCAB

1. Conjecture – An unproven statement based on observations.

Example: Make and test a conjecture about the sign of the product of any three negative integers.

conjecture: The product of any 3 negative integers will be negative.

tests: $-1 \cdot -1 \cdot -1 = -1$ } $-1 \cdot -2 \cdot -3 = -6$ } $-4 \cdot -5 \cdot -6 = -120$
TRUE TRUE TRUE

2. Counterexample – a case when the conjecture is FALSE (proving it false)

Example: Find a counterexample to show that the conjecture is false.

The value of x^2 is always greater than the value of x .

$(1)^2 > 1$ } $(-1)^2 > -1$ } $(.5)^2 > .5$
 $1 > 1$ FALSE } $1 > -1$ TRUE } $.25 > .5$ FALSE

3. Inductive Reasoning –

write a conjecture based on a PATTERN that is observed.

Example: Sketch the next figure in the pattern.



EX]: J, F, M, A, M, ...

J, J

What are the next 2 terms?

4. Deductive Reasoning – uses facts, definition, properties, and theorems, and laws of logic to form a logical argument.